MIL-C-60562A(MU)
20 February 1969
SUPERSEDING
MIL-C-60562(MU)
20 January 1967

MILITARY SPECIFICATION
CASE, CARTRIDGE, 30MM, XM193

1. SCOPE

1.1 This specification covers Case, Cartridge, 30MM, XM193.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-651	- Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1305

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

2.1 STANDARDS (Cont'd)

Military

MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F10542940	- Case, Cartridge, 30MM, XM193
C7643674	- Classification of Cartridge Case Defects, Small Arms Ammunition
D9206805	- Cartridge, 30MM, XM639, Training
F10542500	- Carton, Interplant Shipment 30MM, XM193 Type Cartridge Cases Empty, Unprimed
IEL-10542940	- Inspection Equipment List for: Case, Cartridge, 30MM, XM193

PUBLICATIONS

U.S. Army Materiel Command

AMCR 715-505	- Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge case shall comply with Drawing F10542940, referenced specifications, and other requirements specified herein.

3.2 Manufacturing process.- The cartridge case shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval (see 6.2).

3.3 Hardness.- The cartridge case shall comply with the hardness requirement of Drawing F10542940.

3.4 Residual stress.- The cartridge case shall be free of residual stresses to the degree detectable by the mercurous nitrate test.

3.5 Neutrality.- The cartridge case surfaces shall be free of acids and alkalis.

Source: https://assist.dla.mil -- Downloaded: 20261-23T00:43Z
Check the source to verify that this is the current version before use.

3.6 Function and casualty.- The cartridge case shall function without casualty when assembled into 30MM XM639 type cartridges, conditioned at $70^{o} \pm 5^{o}F.$, and loaded to obtain an average chamber pressure of 32000 psi $\pm$ 2000 psi.

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following.

3.7.1 Metal defects.- The cartridge case shall be free of defects such as metal flaws, folds, wrinkles, splits, cracks, laminations, dents and perforations.

3.7.2 Cleanliness.- The cartridge case shall be free of foreign matter such as corrosion stains, dirt, grease, oil, lint, chips, slivers and shavings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms.

4.2 First article sample. (See 6.4)

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

a. Inspection and non-firing test(s)

100 cartridge cases

b. Firing test(s)

200 cartridge cases

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production (See 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot.- A lot shall consist of cartridge cases produced by one manufactuere in one unchanged process, in accordance with the same drawings and drawing revisions, same specification and specification revision.

4.3.1.2 Lot identification.- Each cartridge case and each packed cartridge case lot shall be identified in accordance with applicable drawings and MIL-STD-1168.

4.3.2 Examination.- Examination for major and minor defects shall be performed as specified herein. Each cartridge case, found to be defective, shall be rejected.

4.3.2.1 Sampling and acceptance criteria.- Sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235, except as otherwise indicated. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered inidividually to determine acceptability. When acceptability is determined on a class basis, in addition to the AQL assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, when three or less defect characteristics are listed in a classification or when sampling procedures of MIL-STD-1235 are used, acceptability shall not be determined on a class basis for major defects (See 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.2.2 AQL's.- The following AQL's are assigned:

a. Individual defect characteristic basis.

Major defects	0.25%
Minor defects	0.40%

b. Defect characteristic, class basis.

Major defects	1.50%
Minor defects	2.50%

c. Individual defect characteristic associated with the class basis.

Major defects	0.40%

4.3.3 Classification of defects 1/.- The classification of defects shall be as follows:

4.3.3.1 Case, Cartridge (see Drawing F10542940).

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Metal defective	Visual
102.	Corrosion	Visual
103.	Oil or grease	Visual
104.	Head configuration (round head or beveled underside)	Visual
105.	Chamfer, head rim, missing	Visual
106.	Profile, maximum	Gage
107.	Total length, maximum	Gage
108.	Diameter formed cylindrical section, min.	Gage
109.	Diameter, middle body taper, minimum	Gage
110.	Length to basic diameter on belt	Gage
111.	Thickness, head rim, min.	Gage
112.	Diameter, rim	Gage
113.	Concentricity, rim diameter on belt	Gage
114.	Diameter, extractor groove, maximum	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.3.1 (Cont'd)

	Defect	Method of Inspection
MAJOR (Cont'd)		
115.	Width, extractor groove, minimum	Gage
116.	Diameter, primer pocket	Gage
117.	Concentricity, primer pocket with body	Gage
118.	Depth, primer pocket	Gage
119.	Diameter, flash hole	Gage
120.	Concentricity, flash hole with primer pocket	Gage
121.	Thickness, wall at .500 inches from inside base, min.	Gage
122.	Diameter, inside mouth, minimum for minimum length, min.	Gage
123.	Diameter, belt	Gage
124.	Concentricity of head rim	Gage
MINOR		
201.	Foreign matter, other than oil, grease or corrosion	Visual
202.	Scratch (other than die scratch)	Visual
203.	Fold, wrinkle or buckle	Visual
204.	Radii or chamfers missing or incorrect	Visual
205.	Total length, minimum	Gage
206.	Thickness, wall, at mouth	Gage
207.	Diameter, extractor groove minimum	Gage
208.	Width, extractor groove, maximum	Gage
209.	Thickness, base	Gage
210.	Diameter, inside mouth, maximum	Gage
211.	Depth, indented head (alternative design)	Gage
212.	Marking, missing, incorrect or illegible	Visual
213.	Surface finish	Visual
214.	Thickness, wall at .500 inches from inside base, max.	Gage

1/ MIL-STD-651 shall apply in defining and evaluating visual defects. No photographs of 30mm cases are included. However, the defect classification of the 20mm M103 Brass Case will serve as a guide for inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.4 Tests.- The tests listed in Table I shall be performed on each cartridge case lot in accordance with the test methods prescribed in 4.4. Sample size and acceptance criteria for each test shall be as specified. Only cartridge cases having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cases shall be thoroughly mixed before being divided into samples for the various tests.

TABLE I

Test		Sample Size	Requirement Paragraph
Hardness	1/	5	3.3
Residual stress	2/	10	3.4
Neutrality	3/	10	3.5
Function and Casualty	4/	200	3.6

1/ Failure of two or more of the sample cartridge cases to comply with the hardness requirements shall cause rejection of the lot. If one of the samples fails to comply, a second sample of 10 cartridge cases shall be randomly selected from the lot and tested. The lot shall be rejected if a total of two or more cartridge cases of the combined first and second samples fail to comply with the applicable hardness requirement. Failure is defined as any individual hardness value obtained that does not comply with the drawing hardness pattern position requirements.

2/ The lot shall be rejected if one or more cartridge cases of the sample split.

3/ The lot shall be rejected if one or more cartridge cases of the sample show evidence of acidity or alkalinity.

4/ The lot shall be rejected if any malfunction or casualty of Table II occurs in number(s) equal to or greater than the applicable "Rej" number. If malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 400 cartridge cases randomly selected from the lot shall be tested. The lot shall be rejected if in the combined samples malfunctions or casualties occur in number(s) equal to or greater than the applicable "Rej" number.

4.3.4.1 Firing defects and associated criteria shall be as specified in Table II. For defect definition, see AMCR 715-505, Vol. 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.4.1 (Cont'd)

TABLE II

Firing Casualties	Acc	Rej
Failure to extract 1/	0	1
Blown primer 2/	0	1
Splits and ruptures 3/		
Splits:		
I	3	8
J	1	6
K, L or M	0	1
Rupture, partial		
J, or K	0	3
L	0	1
Circumferential rupture, complete	0	1
Detached metal 4/	0	1

1/ A fired sample cartridge case remaining in the chamber after the normal extraction cycle will be classed defective, unless it can be shown that the malfunction was caused by a defective weapon.

2/ The sample cartridge case shall be classed defective only if it is evident by visual inspection that both the primer pocket and case head are enlarged and distorted.

3/ Refer to Drawing C7643674 for classifying splits and ruptures resulting from the function and casualty test. If a split or partial rupture extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved shall be used. A sample cartridge case which separates into two or more portions, shall be classed defective as a complete circumferential rupture.

4/ The case lot will not be penalized if malfunction cannot be shown to have resulted from a defective case. There must be evidence of detached metal from fired case exterior. The lot will not be penalized for the detachment of metal shavings from the crimped area of the case interior.

4.3.4.2 Unlisted firing defects.- The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification, occurring in any firing test, indicates that the product is unsuited for the purpose intended.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.5 Packaging and marking inspection.- Inspection for packing and marking shall be in accordance with MIL-STD-1169.

4.3.6 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-10542940.

4.4 Test methods and procedures.

4.4.1 Hardness.- The cases of the test sample(s) will be sectioned as shown on Drawing F10542940. The head sections will have approximately parallel surfaces finished to 63 microinches or finer, care being taken not to change the hardness in obtaining this finish. Four determinations each will be made at the indicated body position for each sample case, two on each half on the interior as shown. Two determinations each will be made at the indicated head position as shown for each sample case. The method of test will be as prescribed in Federal Test Method Standard 151.

4.4.2 Residual stress (Mercurous nitrate).- The method of test shall be as prescribed in AMCR 715-505, Vol. 8 for case mercurous nitrate test.

4.4.3 Neutrality.- The method of testing cartridge cases for acidity and alkalinity will be as follows using red and blue sensitive litmus paper. Neutral distilled water will be used to moisten the paper and the case to provide adhesion. The litmus paper will be handled with forceps to prevent contamination from fingers. Smoking should not be permitted in the vicinity of test since tobacco smoke will turn red litmus blue. Pieces of red and blue sensitivie litmus paper will be placed in several locations both on the interior and exterior of the case. A definite color change is defined as the color produced when blue litmus is immersed for thirty seconds in one thousandth normal (.001N) hydrochloric or sulfuric acid, or when red litmus is immersed for thirty seconds in one thousandth normal (.001N) sodium hydroxide. Litmus paper is considered of sufficient sensitivity if it shows change in color when immersed for 45 seconds, with stirring, in five ten thousandths normal (.005N) solutions of the corresponding acid or alkali mentioned.

4.4.4 Function and casualty.- A propellant charge will be established which will meet the average chamber pressure requirements of 3.6. The two hundred (200) test sample cases will be assembled into cartridges 30mm XM639 Training, Drawing D9206805 and fired in an XM140 Machine Gun in bursts of 50 rounds at a cyclic rate of not less than 400 rounds per minute. The method of test shall be as prescribed in AMCR 715-505, Vol. 8. The gun barrel used will be at ambient temperature at the beginning of the test and will be cooled to ambient temperature between bursts. This test will be conducted at a facility designated by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.- Cartridge cases shall be packaged in accordance with Drawing F10542500.

5.2 Marking and labeling.- Marking on cartons shall be in accordance with MIL-STD-1169.

5.3 Data cards.- Ammunition data cards shall be furnished, and shall comply with requirements of MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Invitation for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 Packing instructions (see 5.1).

6.1.4 Data card requirements (see 5.3).

6.1.5 Provision for the submission and approval of a manufacturing process (see 3.2).

6.1.6 Instructions for shipment of first article inspection and production lot acceptance samples for function and casualty test.

6.1.7 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance test purposes.

6.1.8 Provision for the submission of contractor inspection reports containing final inspection results for each lot of metal parts presented to the Government.

6.2 Process deviation.- A process deviation is defined as a change in the approved basic method of manufacture or an operational change which may alter the metallurgical or physical properties of the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

6.3 AQL's.- The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.4 First article sample firing test.- The firing test of 4.4 will be performed after the sample has been previously accepted by the Government for all other requirements. Additional cartridge cases may be required by the testing facility.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1305- A580

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.